

MUHAMMAD HAMMAD

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Education

Hazara University

Bachelor of Software Engineering

Sep. 2023 – May 2027

Mansehra, KPK

About Me

- Software Engineering student with hands-on experience building production-grade backend APIs using Python, FastAPI, and PostgreSQL. Experienced with Redis caching, async task queues, JWT authentication, and system design fundamentals..

Projects

AI-Powered URL Shortener | *Python, FastApi, PostgreSQL, Redis, SQLAlchemy, ARQ*

January 2026

- * Built a production-grade URL shortening API with two-layer Redis caching reducing database load on repeated requests.
- * Implemented async task queue using ARQ for AI-powered slug generation via Groq, returning responses in under 50ms.
- * Designed JWT authentication, rate limiting per user, and URL expiry with automatic cache invalidation.
- * Used APScheduler for periodic cleanup of expired URLs.

LSC Web App – Two-Sided Marketplace Platform | *Python, Flask, MySQL,HTML5, CSS, JavaScript (Vanilla), Jinja2 Temp*

- * Built a full-stack two-sided marketplace connecting service seekers with local providers across 6 professions including plumbing,electrical, and cleaning services.
- * Implemented role-based access control with three distinct user types (seekers, providers, admins) each with separate dashboards,permissions, and booking workflows.
- * Designed real-time messaging system between customers and service providers with booking status tracking and history.
- * Architected authentication system using Bcrypt password hashing and parameterized SQL queries preventing SQL injection attacks.
- * Built booking engine supporting both direct provider selection and general service requests with status management.

Technical Skills

Languages: Python, SQL, HTML/CSS

Frameworks: FastApi, Flask

Databases: PostgreSQL, MySQL

Tools: Git, GitHub, VS Code, Postman

Technologies: Redis, Docker